

STA2023 – Statistical Methods I
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Chapter 4. Probability

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Learning Objectives:

By the end of this chapter, students should be able to:

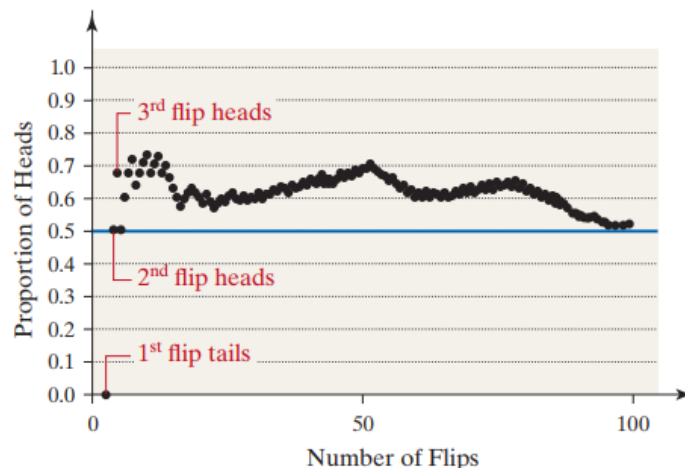
- Understand the random process and apply basic rules of probabilities.
- Compute and interpret probabilities using the empirical, classical, and subjective methods.
- Apply the addition rule for disjoint events and the general addition rule.
- Compute the probability an of event using the Complement Rule
- Identify independent events.
- Use the multiplication rule for independent events and the general multiplication rule.
- Compute at-least probabilities.
- Solve counting problems using the multiplication rule, permutations, and combinations.
- Determine the appropriate probability and counting technique to use.

4.1 Probability Rules

“What’s the chances that happened?!”

- The word “random” suggests an unpredictable result. Predicting random outcomes can be very challenging, such as predicting if a coin will be heads on the next flip. However, if we flipped a coin many times, we would be able to determine the long-term proportion of times a head is observed.
- **Definition:** A random process **refers to a situation where the outcome cannot be predicted with certainty for a single trial.**

- Consider a coin being flipped.
 - On the first flip, we flipped tails (0 heads out of 1). The proportion of heads is $0/1 = 0$.
 - On the second flip, we flipped heads (1 heads out of 2). The proportion of heads is $1/2 = 0.5$.
 - On the third flip, we flipped heads (2 heads out of 3). The proportion of heads is $2/3 = 0.667$.
- This process is then repeated for 100 coin flips.



- After 100 flips, we can see the proportion of heads is just above 0.50.
- What are the chances of flipping a head on this coin?

About 0.5 (the long-run proportion of heads, just above 0.50 after 100 flips). Law of Large Numbers: as the number of trials grows, the observed proportion gets closer to the true probability.

- **Definition: Probability** is a measure of the likelihood of a random event happening. It represents the *long-term proportion* of times a particular outcome is observed if the experiment is repeated many times.

How do we figure out probabilities? How do we apply probabilities? We have some terminology to describe probabilities and the events we apply probabilities to.

- **Definition:** An **experiment** is any process with uncertain results that can be repeated.

Examples:

flipping a coin, rolling a die, drawing a card from a deck, picking an M&M from a bag

- **Definition:** The **sample space, S**, the collection of all possible outcomes of an experiment.

Example:

Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$. Flipping a coin: $S = \{H, T\}$

- **Definition:** An **event** is any collection (subset) of outcomes from the sample space that we are interested in.

Events are usually represented by capital letters like A,B,C,E,F,GA, B, C, E, F, GA,B,C,E,F,G.

A **simple event** is an event with only one outcome.

EXAMPLE 1: A probability experiment consists of rolling a single *fair* six-sided die.

- Identify the outcomes of the probability experiment.
- Determine the sample space.
- Define the event $E = \text{"roll an even number."}$

(a) **Outcomes:** 1, 2, 3, 4, 5, 6

(b) $S = \{1, 2, 3, 4, 5, 6\}$

(c) $E = \{2, 4, 6\}$

EXAMPLE 2: The color of a plain M&M milk chocolate candy can be brown, yellow, red, blue, orange, or green. Suppose a candy is randomly selected from a bag.

- (a) Identify the outcomes of the probability experiment.
- (b) Determine the sample space.
- (c) Define the event E = “selects a red or blue M&M.”

(a) Outcomes: brown, yellow, red, blue, orange, green

(b) $S = \{\text{brown, yellow, red, blue, orange, green}\}$

(c) $E = \{\text{red, blue}\}$

Basic Rules of Probability

Notation:

$P(E)$ = probability of event E . $N(E)$ = # outcomes in E , $N(S)$ = # outcomes in S

Basic Two Rules of Probability:

1. The probability of any event must be between 0 and 1:

$0 \leq P(E) \leq 1$ ($P = 0$ impossible, $P = 1$ certain)

2. The sum of probabilities of **all possible outcomes** in the sample space must equal 1:

$P(e_1) + P(e_2) + \dots + P(e_n) = 1$

A probability model lists:

- All possible outcomes (sample space)
- The probability assigned to each outcome

An **unusual event** is one that has a **low probability** of occurring, usually:

less than 0.05 (5%)

Example:

If there's only a 3% chance of drawing a certain color M&M, that would be considered unusual.

EXAMPLE 3: A probability experiment consists of rolling a single *fair* six-sided die. The probability model is shown below.

Side Rolled	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Verify that the probability model above is a valid probability model.

Each probability is between 0 and 1 ✓ and $1/6 + 1/6 + 1/6 + 1/6 + 1/6 + 1/6 = 6/6 = 1$ ✓ → VALID

EXAMPLE 4: Consider the two probability models below. Determine which probability models are valid.

Event	Probability
A	0.50
B	0.50
C	0

VALID: all between 0 and 1, sum = $0.50 + 0.50 + 0 = 1$ (a probability of 0 is allowed)

Event	Probability
A	0.40
B	0.20

NOT valid: sum = $0.40 + 0.20 = 0.60 \neq 1$

Event	Probability
A	-0.10
B	0.40
C	0.60
D	0.10

NOT valid: -0.10 is negative (probabilities can't be below 0), even though the sum is 1

Methods to Determine Probabilities

Generally, if we want to determine the probability of some event occurring, we have three methods in practice:

- Empirical Method
- Classical Method
- Subjective Method

Empirical Method

- This method relies on relative frequency on data already obtained.
- To approximate probabilities using the empirical approach:

$$P(E) \approx (\text{frequency of } E) \div (\text{number of trials}) = \text{the relative frequency}$$

EXAMPLE 6: The table below represents the results of a survey in which 200 people were asked their means of travel to work.

- Use the survey data to build a probability model for means of travel to work.
- Estimate the probability that a randomly selected individual carpools to work. Interpret the results.
- Would it be unusual to randomly select an individual who walks to work?

Means of Travel	Frequency
Drive alone	83
Carpool	22
Public transportation	10
Walk	5
Other means	3
Work at home	77

(a) Probability = frequency ÷ 200: Drive alone $83/200 = 0.415$ · Carpool $22/200 = 0.11$ · Public $10/200 = 0.05$ · Walk $5/200 = 0.025$ · Other $3/200 = 0.015$ · Work at home $77/200 = 0.385$ (sum = 1 ✓)

(b) $P(\text{carpool}) = 22/200 = 0.11 \rightarrow$ about 11 out of every 100 randomly selected people carpool to work.

(c) $P(\text{walk}) = 0.025 < 0.05 \rightarrow$ YES, unusual.

EXAMPLE 7: Refer back to the M&M example. Suppose that a few bags of M&M's are opened and you record the following results in the table below.

(a) Use the data below to build a probability model for M&M color.

(b) Estimate the probability of selecting a red M&M.

Color	Frequency
Brown	13
Yellow	14
Red	13
Blue	24
Orange	20
Green	16

(a) Total = 13+14+13+24+20+16 = 100

Brown 13/100 = 0.13

Yellow 14/100 = 0.14

Red 13/100 = 0.13

Blue 24/100 = 0.24

Orange 20/100 = 0.20

Green 16/100 = 0.16 (sum = 1 ✓)

(b) P(red) = 13/100 = 0.13

Classical Method

- The classical method does not require a probability experiment to be actually performed. Instead, it relies on counting techniques.
- This method requires *equally likely outcomes*.
- **Using the classical method:**

$P(E) = N(E) \div N(S) = (\text{number of outcomes in } E) \div (\text{total number of equally likely outcomes})$

Example 8: A fair coin is flipped two times. Using the classical method,

- (a) Determine the sample space of the experiment.
- (b) Compute the probability of the event “two heads are flipped.”
- (c) Compute the probability of the event “at least 1 tails is flipped.”

(a) $S = \{HH, HT, TH, TT\}$ $N(S) = 4$

(b) $P(\text{two heads}) = P(HH) = 1/4 = 0.25$

(c) at least 1 tails = $\{HT, TH, TT\} \rightarrow 3/4 = 0.75$ (or $1 - P(HH) = 1 - 1/4 = 3/4$)

Example 9: Suppose that you have two additional tickets to a concert, but you have four potential friends to bring: Adam, Ben, Sam, and Dustin. To be fair, you randomly select two of your friends to go.

- (a) Determine the sample space of the experiment.
- (b) Compute the probability of the event “Adam and Ben go the concert.”
- (c) Compute the probability of the event “Ben goes to the concert.”

(a) $S = \{AB, AS, AD, BS, BD, SD\}$ $N(S) = 6$ (order doesn't matter)

(b) $P(\text{Adam and Ben}) = 1/6 \approx 0.1667$

(c) Ben goes: $\{AB, BS, BD\} \rightarrow P = 3/6 = 1/2 = 0.5$

Subjective Method

Definition: A subjective probability

is a probability based on personal judgment, experience or opinion (an educated guess) — used when the experiment can't be repeated and there is no data.

- Note that this method is a perfectly legitimate method and sometimes is the *only* method of assigning probabilities to events.
- For example, an economist may be asked to predict the likelihood the economy will fall into recession next year. We can't conduct an experiment "n" times and count the number of years in which the economy fell into a recession.

4.2 The Addition Rule and Complements

- In the upcoming sections, we generally care about two events. There are different probability rules depending on how these two events are related.
- **Definition:** Two events are **disjoint** if **they have NO outcomes in common — they cannot both happen at the same time (also called mutually exclusive). Venn diagram: the two circles do not overlap.**
- We can use **Venn diagrams** to show the relationship between two events.

Example 10: Suppose we randomly select chips from a bag. Each chip is labeled 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Let E represent the event "choose a number less than or equal to 2," and let F represent the event "choose a number greater than or equal to 8."

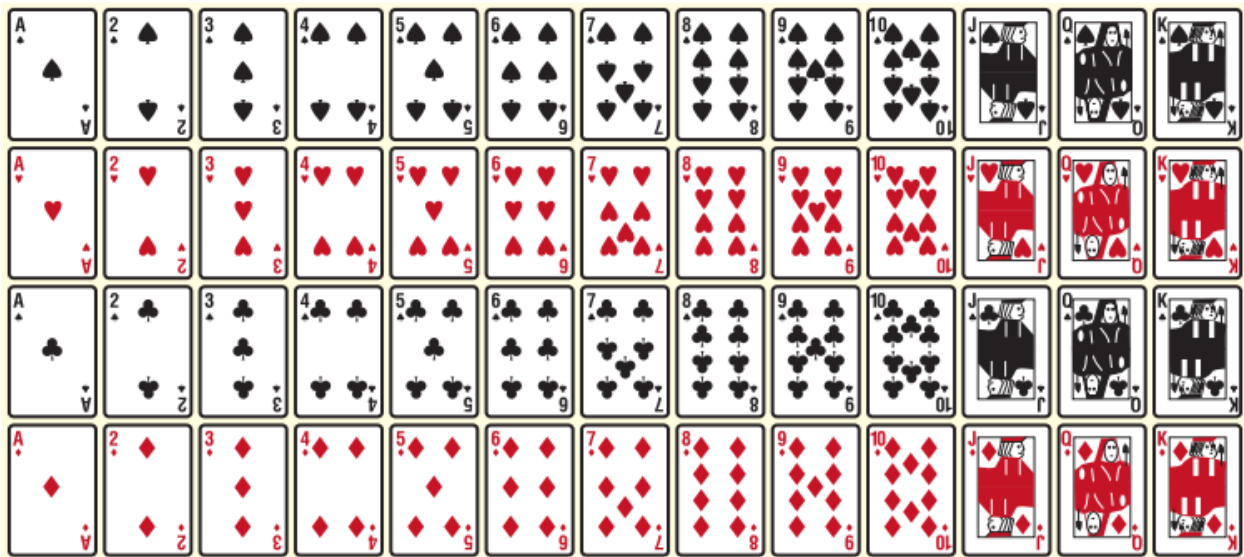
**$E = \{0, 1, 2\}$, $F = \{8, 9\}$. No outcome in common $\rightarrow$ E and F are DISJOINT.
 $P(E) = 3/10$, $P(F) = 2/10$, $P(E \text{ or } F) = 3/10 + 2/10 = 5/10 = 0.5$**

Addition Rule for Disjoint Events

Definition: If E and F are disjoint events, then

$$\mathbf{P(E \text{ or } F) = P(E) + P(F)} \quad \text{(for 3 disjoint events: } \mathbf{P(E \text{ or } F \text{ or } G) = P(E) + P(F) + P(G)})$$

Example 11: Suppose that a single card is selected from a standard 52-card deck, such as the one shown below:



- (a) Compute the probability of the event E = “drawing a king.”
- (b) Compute the probability of the event E = “drawing a king” or F = “drawing a queen” or G = “drawing a jack.”
- (c) Compute the probability of the event A = “drawing a heart” or B = “drawing a diamond.”

(a) $P(\text{king}) = 4/52 = 1/13 \approx 0.0769$

(b) $P(\text{king or queen or jack}) = 4/52 + 4/52 + 4/52 = 12/52 = 3/13 \approx 0.2308$ (disjoint)

(c) $P(\text{heart or diamond}) = 13/52 + 13/52 = 26/52 = 1/2 = 0.5$

The General Addition Rule

- What happens if we want to find $P(E \text{ and } F)$, but they aren't disjoint?

Suppose we are randomly selecting chips from a bag. Each chip is labeled 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9. Let E represent the event “choose an odd number,” and let F represent the event “choose a number less than or equal to 4.” Because the events have the outcomes 1 and 3 in common, the events are not disjoint.

$E = \{1, 3, 5, 7, 9\}$ $P(E) = 5/10$. $F = \{0, 1, 2, 3, 4\}$ $P(F) = 5/10$. $E \text{ and } F = \{1, 3\}$ $P(E \text{ and } F) = 2/10$.

Adding $P(E) + P(F)$ counts 1 and 3 twice, so subtract the overlap:

$P(E \text{ or } F) = 5/10 + 5/10 - 2/10 = 8/10 = 0.8$ (check: $E \text{ or } F = \{0, 1, 2, 3, 4, 5, 7, 9\} = 8 \text{ outcomes } \checkmark$)

Definition: The general Addition Rule

For any two events E and F :

$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$

(if E and F are disjoint, $P(E \text{ and } F) = 0$ and it becomes the disjoint rule)

Example 12: Suppose a single card is selected from a standard 52-card deck. Compute the probability of the event $E = \text{“drawing a king”}$ or $F = \text{“drawing a diamond.”}$

$$P(\text{king or diamond}) = P(\text{king}) + P(\text{diamond}) - P(\text{king of diamonds}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13 \approx 0.3077$$

Probabilities can also be calculated from **contingency tables** which relate two categories of data. There is a **row variable** and a **column variable**.

Example 13: Consider the table below represent the marital status of males and females 15 years old or older in the US in 2021.

	Males (in millions)	Females (in millions)
Never married	49.5	43.2
Married	66.3	64.2
Widowed	3.5	11.5
Divorced	12.3	16.6
Separated	2.0	2.6

Totals: males 133.6, females 138.1, grand total 271.7. Widowed row 15.0, divorced row 28.9.
 (a) $133.6/271.7 \approx 0.4917$ (b) $15.0/271.7 \approx 0.0552$ (c) $(15.0 + 28.9)/271.7 = 43.9/271.7 \approx 0.1616$
 (d) $P(\text{male}) + P(\text{widowed}) - P(\text{male and widowed}) = (133.6 + 15.0 - 3.5)/271.7 = 145.1/271.7 \approx 0.5340$

- Determine the probability that a randomly selected US resident 15 years old or older is male.
- Determine the probability that a randomly selected U.S. resident 15 years old or older is widowed.
- Determine the probability that a randomly selected U.S. resident 15 years old or older is widowed or divorced.
- Determine the probability that a randomly selected U.S. resident 15 years old or older is male or widowed.

Complement of an Event

Sometimes we want to find the probability of an event *not* occurring.

Definition: Let S denote the sample space of a probability experiment and let E denote an event. The **complement of E** ,

written E^c (or E' or “not E ”), is all outcomes in the sample space that are NOT in E .

Complement Rule: $P(E^c) = 1 - P(E)$ so $P(E) + P(E^c) = 1$

Illustration of the Complement:

Rectangle = sample space S ; circle inside = E ; everything in the rectangle outside the circle = E^c

Example 14: According to the National Gambling Impact Study Commission, 52% of Americans have played state lotteries. What is the probability that a randomly selected American has not played a state lottery?

$P(\text{not played}) = 1 - P(\text{played}) = 1 - 0.52 = 0.48$

4.3 Independence and the Multiplication Rule

- In last section, we often knew the probability of “E and F” to find the probability of “E or F.” In this section, we will focus on if the probability of “E and F” is unknown.
 - When calculating probabilities, it is useful if we know that two events are “independent.”
 - **Definition:** Two events E and F are **independent** if **the occurrence of E does NOT affect the probability of F (and vice versa). If it does, the events are DEPENDENT.**
-
- Think about flipping a fair coin twice.
 - Does the fact that you obtained a head on the first toss have any effect on the likelihood of obtaining a head on the second toss?
 - It doesn't so these two events would be considered independent.
 - Think about rolling a six-sided die.
 - What is the probability of rolling a 6? → **1/6**
 - What if you know you rolled an even number? → **{2,4,6} → 1/3**
 - Since the probability of rolling 6 changes once you know that you rolled a 4 or higher, these events would be considered dependent.

Example 15:

- (a) Suppose you flip a coin and roll a die. Are the events “obtain a head” and “roll a 5” independent or dependent?
- (b) Would you expect the events “earned a bachelor's degree” and “earn more than \$100,000 per year” independent?
- (c) Two 24-year-old male drivers who live in the United States are randomly selected. Are the events “male 1 gets in a car accident during the year” and “male 2 gets in a car accident during the year” considered independent or dependent?

Example 15 answers: (a) Independent — the coin can't affect the die. (b) Dependent — a degree changes the chance of earning over \$100,000. (c) Independent — one driver's accident doesn't affect the other's.

Multiplication Rule for Independent Events

- Suppose that you flip a fair coin twice. What is the probability that you obtain a head on both flips?

- If you listed all events in the sample space:

$$S = \{HH, HT, TH, TT\} \rightarrow P(HH) = 1/4$$

- What if we separated it out into two separate events:

$$P(H \text{ on 1st}) \times P(H \text{ on 2nd}) = 1/2 \times 1/2 = 1/4 \quad (\text{same answer})$$

Definition: If E and F are independent events, then

$$P(E \text{ and } F) = P(E) \cdot P(F) \quad (\text{and } P(E \text{ and } F \text{ and } G) = P(E) \cdot P(F) \cdot P(G), \dots)$$

Example 16: The probability that a randomly selected 24-year-old male will survive the year is 0.9986 according to the *National Vital Statistics Report*, Vol. 56, No. 9.

- (a) What is the probability that two randomly selected 24-year-old males will survive the year?
- (b) What is the probability that three randomly selected 24-year-old males will survive the year?
- (c) What is the probability that twenty randomly selected 24-year-old males will survive the year?

$$(a) 0.9986 \times 0.9986 = 0.9986^2 \approx 0.9972$$

$$(b) 0.9986^3 \approx 0.9958$$

$$(c) 0.9986^{20} \approx 0.9724$$

Computing At-Least Probabilities

- The phrase “at least” means “greater than or equal to.”
- When calculating “at least” probabilities, it’s often easier to use the rule of complements and set $P(\text{at least } 1) = 1 - P(\text{none})$. We will use an example to illustrate.

Example 17: Compute the probability that at least one male out of 1000 aged 24 years will die during the course of the year if the probability that a randomly selected 24-year-old male survives the year is 0.9986.

$$\begin{aligned} P(\text{at least one dies}) &= 1 - P(\text{none die}) = 1 - P(\text{all 1000 survive}) \\ &= 1 - 0.9986^{1000} \approx 1 - 0.2464 = 0.7536 \end{aligned}$$

4.4 Conditional Probability and the General Multiplication Rule

- In the last section, all probability calculations relied on events being independent. What happens if they’re dependent?
- For example, what’s the probability of getting in an accident if you drive to class? Well, that might depend on the weather conditions. It might be higher during a rainstorm than on a sunny day.
 - For example, the National Highway Safety Administration states that the probability of a randomly selected vehicle crash has a fatality is 0.0054.
 - However, among crashes which the driver is drowsy, that probability increases to 0.0196.
- If you are given some sort of information about an event occurring, then we can find the **conditional probability** of another event occurring.
- **Definition:**

Conditional probability $P(F | E)$ = the probability that F occurs GIVEN that E has already occurred (read “ F given E ”). The given information shrinks the sample space to only the outcomes where E happened.

Example 18: Suppose a single six-sided die is thrown. What is the probability that the die comes up a six? Now suppose that the die is rolled a second time, but we are told it was an even number. What is the probability that the die comes up a six?

P(six) = 1/6. Given even → sample space is now {2, 4, 6} → P(six | even) = 1/3

Conditional Probability Rule

Refer to the example relating sex to marital status:

	Males (in millions)	Females (in millions)	Totals (in millions)
Never married	49.5	43.2	92.7
Married	66.3	64.2	130.5
Widowed	3.5	11.5	15.0
Divorced	12.3	16.6	28.9
Separated	2.0	2.6	4.6
Totals	133.6	138.1	271.7

If we randomly selected someone, what's the probability that they are widowed?

P(widowed) = 15.0/271.7 ≈ 0.0552

Suppose we know that the individual is female. How does that change the probability? Now, the sample space only includes females.

P(widowed | female) = 11.5/138.1 ≈ 0.0833 (divide by the female total only)

Definition: Conditional Probability Rule

If E and F are any two events, then:

$$P(F | E) = P(E \text{ and } F) \div P(E) = N(E \text{ and } F) \div N(E)$$

(in a table: the cell count $\div$ the total of the row or column named after “given”)

Example 19: Refer back to the marital status example with the data below:

	Males (in millions)	Females (in millions)	Totals (in millions)
Never married	49.5	43.2	92.7
Married	66.3	64.2	130.5
Widowed	3.5	11.5	15.0
Divorced	12.3	16.6	28.9
Separated	2.0	2.6	4.6
Totals	133.6	138.1	271.7

- (a) Compute the probability that a randomly selected individual has never married, given they are male.

$$P(\text{never married} | \text{male}) = N(\text{never married and male}) \div N(\text{male}) = 49.5 \div 133.6 \approx 0.3705$$

- (b) Compute the probability that a randomly selected individual is male, given the individual has never married.

$$P(\text{male} | \text{never married}) = N(\text{male and never married}) \div N(\text{never married}) = 49.5 \div (49.5 + 43.2) = 49.5/92.7 \approx 0.5340$$

General Multiplication Rule

- Sometimes we know a conditional probability, but we want to find the probability of event E and event F occurring.

- **Definition: General Multiplication Rule**

The probability that two events E and F both occur is

$P(E \text{ and } F) = P(E) \cdot P(F | E)$ (works whether or not the events are independent)

Example 20: The probability that a driver who is speeding gets pulled over is 0.8. The probability that a driver gets a ticket, given that they are pulled over, is 0.9. What is the probability that a randomly selected driver who is speeding gets pulled over and gets a ticket?

$P(\text{pulled over and ticket}) = P(\text{pulled over}) \times P(\text{ticket} | \text{pulled over}) = 0.8 \times 0.9 = 0.72$

Determining Independence

- Notice how parts (b) and (c) were close in value, regardless if you assumed independence or not.
- If small random samples are taken from large populations without replacement, it is reasonable to assume independence of the events. As a rule of thumb, if the sample size is less than 5% of the population size, we treat the events as independent.

Definition: Two events E and F are independent if

$P(F | E) = P(F)$ (equivalently $P(E | F) = P(E)$). If the two probabilities are NOT equal $\rightarrow$ dependent.

Example 21: Refer back to the marital status example with the data below:

	Males (in millions)	Females (in millions)	Totals (in millions)
Never married	49.5	43.2	92.7
Married	66.3	64.2	130.5
Widowed	3.5	11.5	15.0
Divorced	12.3	16.6	28.9
Separated	2.0	2.6	4.6
Totals	133.6	138.1	271.7

If we randomly select an individual, are the events “widowed” and “female” independent?

$P(\text{widowed}) = 15.0/271.7 \approx 0.0552$ but $P(\text{widowed} | \text{female}) = 11.5/138.1 \approx 0.0833$

$0.0833 \neq 0.0552 \rightarrow$ NOT independent (dependent): knowing the person is female raises the chance of being widowed.

4.5 Counting Techniques

- While counting techniques sounds like a simple concept, counting plays a major role in many diverse areas, including probability. Simply counting the number of ways you can do something can be deceptively difficult.
 - In fact, there's a whole class about counting at UCF: **MAD 4203 Introduction to Combinatorics** (it has a prerequisite of calculus 2 and logic and proofs in mathematics, the course is no joke!)
- We will cover a few basic methods for probability calculations.

General Counting Principle

- Suppose you are at a fancy restaurant that gives you the following options for an appetizer, entrée, and dessert:
 - Appetizer
 - Soup
 - Salad
 - Entrée
 - Baked Chicken
 - Roasted beef au jus
 - Broiled beef patty
 - Dessert
 - Ice Cream
 - Cheesecake

2 appetizers × 3 entrées × 2 desserts = 12 different meals

Definition: Multiplication Rule of Counting

If a task consists of a sequence of choices in which there are p selections for the first choice, q selections for the second choice, r selection for the third choice, and so on, then

the number of different ways to complete the task is $p \cdot q \cdot r \cdot \dots$

Example 22: The International Air Transport Association (IATA) assigns three-letter codes to represent airport locations. For example, the code for Fort Lauderdale International Airport is FLL. How many different airport codes are possible (assuming repeated letters are allowed)?

$26 \times 26 \times 26 = 26^3 = 17,576$ codes

Example 23: Three members from a 14-member committee are to be randomly selected to serve as chair, vice-chair, and secretary. The first person selected is the chair, the second is the vice-chair, and the third is the secretary. How many different committee structures are possible?

$14 \times 13 \times 12 = 2,184$ (one fewer choice each time — no repeats)

- A symbol you may see before in math is the factorial symbol, or “!”. In counting, we often have to use factorials.
- **Definition:** If $n \geq 0$ is an integer (whole number), the **factorial symbol, $n!$** , is defined as:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 3 \cdot 2 \cdot 1, \text{ and } 0! = 1 \quad (\text{e.g. } 5! = 120)$$

Example 24: You have just been hired as a book representative. On your first day, you must travel to seven schools to introduce yourself. How many different routes are possible?

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5,040 \text{ routes}$$

Permutations

- Examples 28 and 29 above are special cases of the multiplication rule of counting and are actually easier solved using permutations.
- **Definition:** A **permutation** is an ordered arrangement in which r objects are chosen from n distinct objects (such that $r \leq n$). Repetition is not allowed (the same object cannot be selected twice). We use the following symbols and formula:

$${}_nP_r = n! \div (n - r)! \quad (\text{order MATTERS, no repeats})$$

Example 25: Evaluate ${}_7P_5$ and ${}_5P_5$.

$${}_7P_5 = 7! \div (7 - 5)! = 7! \div 2! = 5040 \div 2 = 2,520$$

$${}_5P_5 = 5! \div 0! = 120 \div 1 = 120$$

Using Excel:

=PERMUT(n, r)

In the examples above,

=PERMUT(7, 5)

=PERMUT(5, 5)

Example 26: Consider a race with 10 runners. How many ways can 10 runners finish first, second, and third?

$${}_{10}P_3 = 10! \div 7! = 10 \cdot 9 \cdot 8 = 720$$

Combinations

- In permutations, order mattered (so selecting ABC is different than BAC).
- Sometimes, we don't care about order. For example, in poker, we don't care about the order in which the cards are received.

Definition: A **combination** is a collection, without regard to order, in which r objects are chosen from distinct objects (with $r \leq n$). We use the following notation and formula:

$${}^nC_r = n! \div [r! (n - r)!] \quad (\text{order does NOT matter})$$

In Excel:

=COMBIN(n, r)

Example 27: Roger, Ken, Tom, and Jay are going to play golf. They will randomly select teams of two players each. How many ways can teams be selected? List all possible combinations, then use the combinations formula.

List: RK, RT, RJ, KT, KJ, TJ → 6 teams

Formula: ${}^4C_2 = 4! \div (2! \cdot 2!) = 24 \div (2 \cdot 2) = 6$

Example 28: How many different simple random samples of size 4 can be obtained from a population whose size is 20?

$${}^{20}C_4 = 20! \div (4! \cdot 16!) = (20 \cdot 19 \cdot 18 \cdot 17) \div (4 \cdot 3 \cdot 2 \cdot 1) = 116,280 \div 24 = 4,845$$

Using Counting Methods in Probability Calculations

- In probability, we can use counting to count the total number of outcomes for a certain probability experiment. We can count the number of outcomes in the numerator and then divide by the total number of outcomes.

Example 29: Consider a standard deck of 52 cards. You are dealt 4 cards. What is the probability that all 4 cards match? Solve this by answering the questions below.

- (a) How many ways can 4 cards be selected from a 52-card deck?
- (b) Of all the possible hands listed in part (a), how many of them have all 4 cards that match?
- (c) What is the probability that all 4 cards match?

(a) ${}^{52}C_4 = 52! \div (4! \cdot 48!) = (52 \cdot 51 \cdot 50 \cdot 49) \div 24 = 270,725$ hands

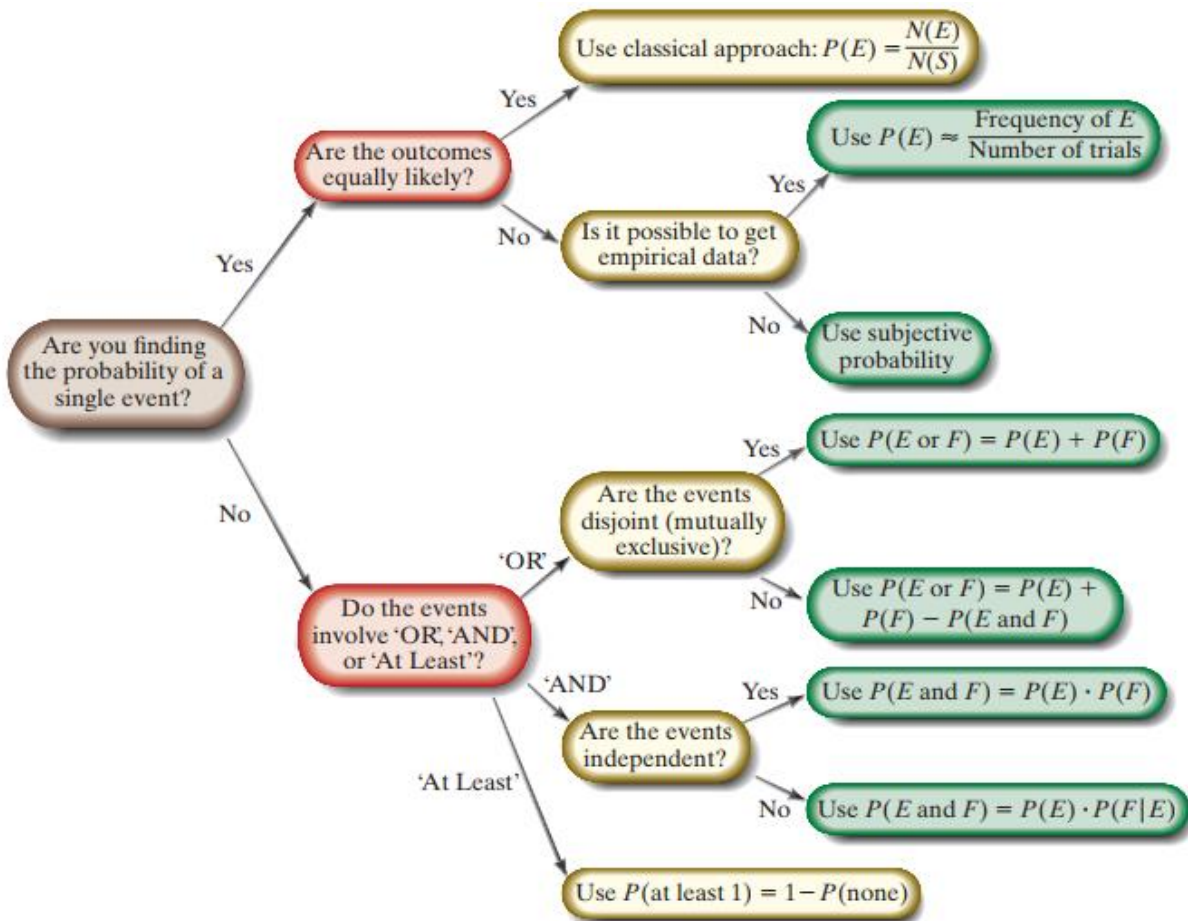
(b) One four-of-a-kind for each of the 13 ranks $\rightarrow$ 13 hands

(c) $P = 13 \div 270,725 \approx 0.00005$

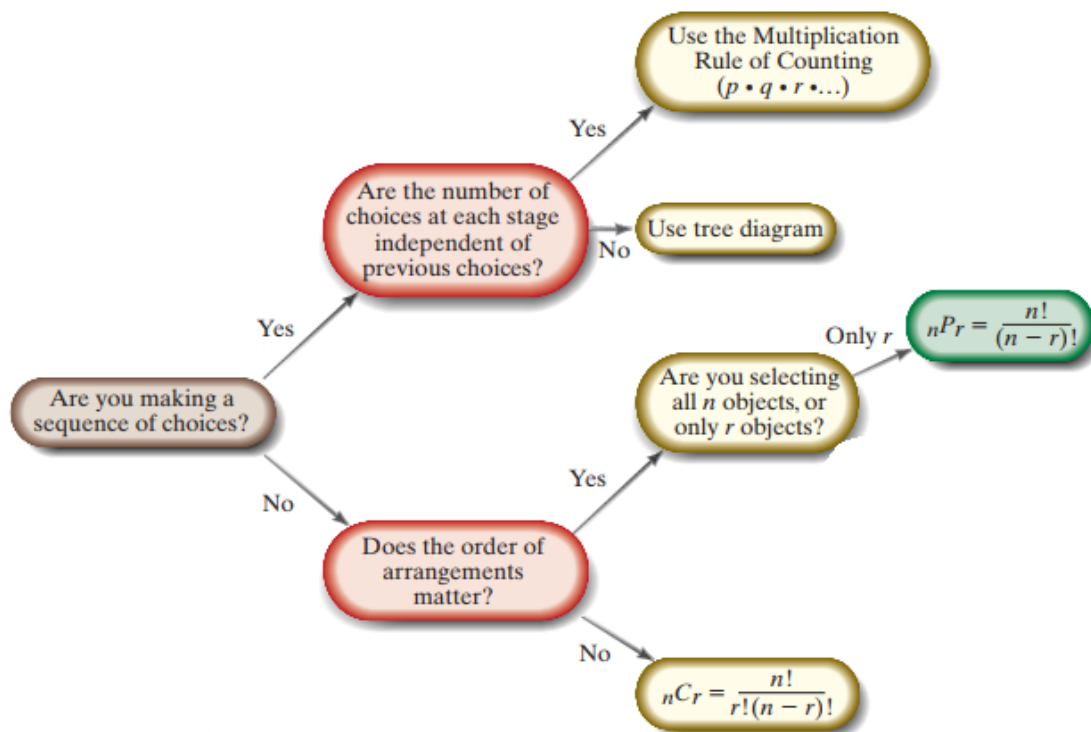
- How you know what method to use?
 - If you are making a sequence of choices: use Multiplication Rule of Counting
 - If you are choosing r objects of a total of n objects:
 - If order matters, use permutation.
 - If order doesn't matter, use combination.

4.7 Determining Which Method to Use

- We just covered a lot of probability rules and counting techniques. When you go to solve a problem, often times you will not be directly given the technique you should use. How do you determine that?
- Often times, there are multiple ways to the correct answer, but certain methods are easier than other. You can use the flow diagram below to determine what rule should be used.



- What about counting techniques?



Note: there are other counting techniques discussed in 5.5, but these are just the ones we covered.

Recall that for Excel:

- Permutations
 - =PERMUT(n, r)
- Combinations
 - =COMBIN(n, r)

Formulas

Empirical Probability

$$P(E) \approx \frac{\text{frequency of } E}{\text{number of trials of experiment}}$$

Classical Probability

$$P(E) = \frac{\text{number of ways that } E \text{ can occur}}{\text{number of possible outcomes}} = \frac{N(E)}{N(S)}$$

Addition Rule for Disjoint Events

$$P(E \text{ or } F) = P(E) + P(F)$$

General Addition Rule

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$$

Probabilities of Complements

$$P(E^c) = 1 - P(E)$$

Multiplication Rule for Independent Events

$$P(E \text{ and } F) = P(E) \cdot P(F)$$

Multiplication Rule for n Independent Events

$$P(E_1 \text{ and } E_2 \text{ and } E_3 \cdots \text{ and } E_n) = P(E_1) \cdot P(E_2) \cdot \cdots \cdot P(E_n)$$

Conditional Probability Rule

$$P(F|E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$$

General Multiplication Rule

$$P(E \text{ and } F) = P(E) \cdot P(F|E)$$

Factorial Notation

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \cdots \cdot 3 \cdot 2 \cdot 1$$

Combination

$${}_nC_r = \frac{n!}{r!(n - r)!}$$

Permutation

$${}_nP_r = \frac{n!}{(n - r)!}$$